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Basic Analysis of Walking with Coffee in A Cup

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Abstract

To evaluate the factors that influence the spillage of coffee in a cup, we mainly utilize the knowledge of liquid sloshing in computational fluid dynamics. We construct our model mostly based on the Navier-Stokes equations that describe the motion of fluid substances. The result of our model is a function that specifies the surface elevation above the initial water level. In developing our model, we derive a system of partial differential equations with Lagrangian method. Then, we solve the PDEs by using MATLAB and acquire the graphs of the model. Also, based on the equations we solve earlier in the model, we yield another function to describe surface elevation with more detailed graphs of the movement.

Then we numerically analyze factors that affect surface elevation according to the model. By plotting graphs of comparing different states, we get the relationship of how these factors operate.

In Task two, we simulate the effect of vibration by setting an entrapment for the robot and observe the surface elevation of coffee in the cup. With experimental results from literature, we acquire the descriptive result of the impact.

Introduction

During office break, holding the freshly made coffee in the hallway might be the greatest pleasure of the day. Nevertheless, there would always have some careless coworkers to run into your cup of coffee. To avoid the chance of getting the coffee spilled, we turn to the liquid sloshing engineering of fluid dynamics.

In fluid dynamics, slosh refers to the movement of liquid inside an object. More specifically, the liquid must have a free surface to constitute a slosh dynamics problem, where the dynamics of the liquid can interact with the container to alter the system dynamics significantly. In our case, the cup is a cylinder container and coffee makes the perfect free-surface liquid. Liquid sloshing engineering is mostly applied in the astronomy, which includes the propellant slosh in spacecraft tanks and rockets, and cargo slosh in ships and trucks transporting liquids.

There have been a lot of methods to approach the sloshing problem. The most classic is the Navier-Stokes equations, which describe the motion of fluid substances. We will implement Navier-Stokes equations to derive our mathematical model and control the sloshing of coffee.

Assumptions

- The fluid is assumed to be homogenous, continuous, isotropic, inviscid and Newtonian and exhibits only limited compressibility.
- The interior of the cup is frictionless and the contact of the coffee and the cup will not affect the liquid motion.
- The coffee cup is abstracted to be a perfect cylinder container, without any concave or convex part.
- To simplify our model, we ignore the influence of airflow and mainly focus on the liquid motion itself and the vibrations.
- Our robot is moving at a constant speed in a straight path with the excitation at the beginning.
- The robot is highly continuous and we assume the movement is not affected by its height, weight or other human factors.
- The temperature in the experimental environment is constant and will not affect the liquid.

- The vibrations obviating the progress of robot are assumed to be random.
- The excitation is assumed to be harmonic, after the motion was started from the rest.

Terminology and Definition

$\eta(t, x)$: surface elevation above the point $(x, 0)$

ω_n : angular velocity

Task1 The Model

Model Overview

There are various movements of the containing liquid. Typical example is horizontal movement along a straight or curved path, vertical movement, rotations or combinations of these. Here, we only study the horizontal motion along a straight path combined with rotation in the vertical plane aligned with the movement of direction.

Initially, we only study the horizontal movement without rotation and will add the rotational acceleration later. In the first section, we divide our model into two parts. We govern the the flow in the fluid in the first part and then the fluid flow in a horizontally accelerated open container. The model is basically using the Lagrangian method.

Modeling of Slosh

Within the fluid

According to the Navier-Stokes equations, we examine the fluid motion within the fluid. The first equation describes the conservation of mass

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho v) = 0 \quad (1.1)$$

Table 1.
Model Parameters

Parameter	Meaning
ρ	the density of the fluid
$\mathbf{v} = (v_x, v_y, v_z)$	flow velocity vector field in the fluid
p	pressure
$\mathbf{f} = (f_x, f_y, f_z)$	the external force vector field
μ	the viscosity of liquid
λ	the factor of volume compression
ω	the vorticity of the flow
ϕ	potential of the potential flows
h	liquid depth
a	the diameter of the liquid in the cylinder container
$\mathbf{g}$	gravity acceleration
$\mathbf{u}(t)$	the horizontal acceleration of the container

We have four unknowns in the equation of the conservation of mass. Unless the density of liquid is given and the problem is single dimensional, we are not able to solve this equation without more equations. Therefore, we have the conservation of momentum

$$\rho \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla p + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{v}) + \lambda \nabla^2 \mathbf{v} + \rho \mathbf{f} \quad (1.2)$$

The equations (1.1) and (1.2) form the Navier-Stokes equations. In our case, we have assumed that the liquid is incompressible ($\lambda = 0$) and inviscid ($\mu = 0$). Also, we assume that the force field has a potential that excludes forces generated by rotational acceleration of the container. Thus, with the conditions above, we solve the Navier-Stokes equations to give

$$\frac{\partial \phi}{\partial t} + \frac{1}{2} |\nabla \phi|^2 + \frac{p}{\rho} + V = C(t) \quad (1.3)$$

where $C(t)$ is an arbitrary function of time. This equation is known as the Bernoulli's equation.

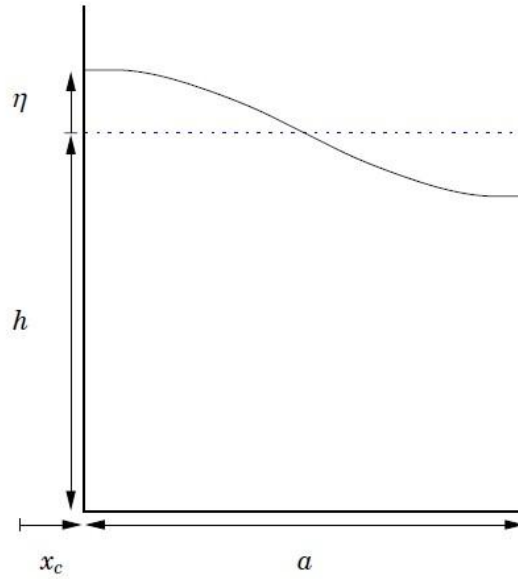


Figure 1 Cross section of the container

On the Fluid Boundary

In this problem, the boundary is considered to be the walls of the coffee cup, bottom and the free surface of the fluid. The velocity at the wall is zero on the walls and bottom in the normal direction of the wall. On the fluid surface $z = \eta(t, x, y)$ there are two boundary conditions: one dynamic and one kinematic.

The dynamic condition can be expressed with Bernoulli's equation

$$\frac{\partial \phi(t, x, y, \eta)}{\partial t} + \frac{1}{2} |\nabla \phi(t, x, y, \eta)|^2 + \frac{p(t, x, y, \eta)}{\rho} + V(t, x, y, \eta) = C(t) \quad (1.4)$$

The kinematic condition is given by

$$\frac{\partial \eta}{\partial t} + \frac{\partial \phi}{\partial z} \frac{\partial \eta}{\partial x} + \frac{\partial \phi}{\partial y} \frac{\partial \eta}{\partial y} = \frac{\partial \phi}{\partial z} \quad (1.5)$$

With a horizontal acceleration in the cup

Since the liquid in the cup is further assumed to be irrotational, the surface elevation above the point $(x, 0)$ is given by $\eta(t, x)$. After separate analysis of

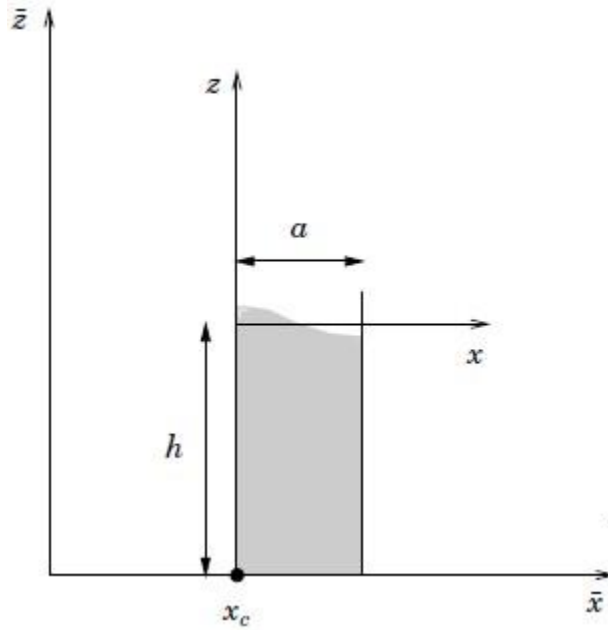


Figure 2 Illustration of the coordinate system

horizontal and vertical acceleration, we give the following description of the fluid flow in the container

$$\nabla^2 \phi(t, x, z) = 0 \quad (1.6)$$

$$\frac{\partial \phi(t, x, z)}{\partial t} + \frac{p(t, x, z)}{\rho} + \frac{1}{2} |\nabla \phi(t, x, z)|^2 + gz + u(t)x = C(t) \quad (1.7)$$

$$\frac{\partial \eta(t, x)}{\partial t} + \frac{\partial \phi(t, x, z)}{\partial x} \frac{\partial \eta(t, x)}{\partial x} = \frac{\partial \phi(t, x, z)}{\partial z} \quad (1.8)$$

$$\frac{\partial \phi(t, 0, z)}{\partial x} = 0, \frac{\partial \phi(t, a, z)}{\partial x} = 0, \frac{\partial \phi(t, x, -h)}{\partial z} = 0 \quad (1.9)$$

Using the method of the separation of variables and with the assumption that pressure is equal everywhere on the surface, we have

$$\omega_n = \sqrt{\frac{n\pi g}{a} \tanh \frac{n\pi h}{a}} \quad (1.10)$$

$$\eta(t, x_m) = \frac{4a}{g\pi^2} \left(\sum_{n=1,3,5}^{\infty} \frac{1}{n^2} \cos \frac{n\pi x_m}{a} \frac{\omega_n^2}{p^2 + \omega_n^2} \right) u(t) \quad (1.11)$$

From the model (1.10) and (1.11), we could see that the elevation of surface is related to the diameter of the container and the depth of liquid, also the pressure and implemented horizontal acceleration. This result matches our daily observation as well. With more depth and smaller diameter, the angular velocity is likely to be larger.

Since the coffee cup is open on the top, tangential stress is negligible due to the relatively larger fluid density comparing with the air and accordingly the only stress at such surface is the normal pressure. The summation of the forces normal to free surface must be balanced by the atmospheric pressure. In the study of Celebi and Akyldiz, we have

$$\frac{dp}{d\rho} = c^2$$

where c is the adiabatic speed of sound.

The horizontal acceleration $u(t)$ stimulates the container to move. In this case, we make the function to be exponential, i.e.

$$u(t) = \alpha t^\beta$$

where α, β are the parameters to control the motion of the cup.

Analytic Approach

In the study of Miles[1976], we solve the flow problem based on the Lagrange mechanics. The fluid is modeled as a potential flow by the Laplace equation. Using the separation of parameters, we have

$$\eta(t, x_m) = \sum_{n=1}^{\infty} q_n(t) \Psi_n(x_m), \Psi_n(x) = \sqrt{2} \cos \frac{n\pi}{a} x \quad (1.12)$$

The function $q_k(t)$ are given by the following nonlinear differential algebraic equations (DAE)

$$\sum_{m=1}^{\infty} (a_{mn} q_m + \sum_{l=1}^{\infty} b_{lmn} q_l q_m) + g q_n = c_n u, n = 1, 2, \dots \quad (1.13)$$

where a_{mn} and b_{lmn} are functions of $q = (q_1, \dots, q_\infty)$ and u is the applied horizontal acceleration. The coefficients are given by

$$a_{mn} = \delta_{mn} a_n + \sum_{l=1}^{\infty} (a_{lmn} q_l + \frac{1}{2} \sum_{j=1}^{\infty} a_{jlmn} q_j q_l + \dots)$$

$$b_{lmn} = \frac{1}{2} \left(\frac{\partial a_{mn}}{\partial q_l} + \frac{\partial a_{nl}}{\partial q_m} - \frac{\partial a_{lm}}{\partial q_n} \right)$$

$$c_n = -\frac{1}{a} \int_0^a x \psi_n dx = a \sqrt{2} \frac{1 - (-1)^n}{n^2 \pi^2}$$

with

$$\delta_{mn} = \begin{cases} 1, & m = n \\ 0, & m \neq n \end{cases}$$

$$a_n = \frac{g}{\omega_n^2}, \omega_n = \sqrt{\frac{n\pi g}{a}} \tanh \frac{n\pi h}{a}$$

$$a_{lmn} = C_{lmn} - D_{lmn} a_m a_n$$

$$a_{jlmn} = -D_{jlmn} (a_m + a_n) + 2 \sum_{i=1}^{\infty} D_{jmi} D_{lni} a_i a_m a_n$$

The models given in (1.11), (1.12) and (1.13) are based on separation of variables and describes the surface shape by

$$S(t, x) = \sum_{n=1}^{\infty} c_n(t) \cos \frac{n\pi}{a} x \quad (1.14)$$

where $S(t, x)$ is the surface elevation at time t in the horizontal position x relative to the cup. The behavior of surface movement is then given by $c_n(t)$. In our mode, we assume the time interval is $h_t = 20\text{ms}$ and the space interval is $h_s = 0.15\text{mm}$. The average coffee cup is 0.0762m wide which gives 508 data points and the duration of the simulation is 20s which gives 1000 data points. Therefore, we could replace the parameters and have $S(kh_t, lh_s)$ with $k=0,1,\dots,999$ and $l=0,1,\dots,507$.

The coefficients are calculated by using the discrete approximation of the integral $c_n(t) = \frac{2}{a} \int_0^a s(t, x) \cos \frac{n\pi x}{a} dx$

$$c_n(kh_t) = \frac{2}{508} \sum_{l=0}^{507} S(kh_t, lh_s) \cos \frac{n\pi l}{507} \quad (1.15)$$

Numerical Computation

Parameters

First of all, we determine some parameters of the model.

Average coffee cup is approximately with the volume of 12 oz and the mug is of the size both 3 1/2 feet wide and 3 1/2 feet deep. In our problem, to facilitate calculation, we assume the depth of coffee is 10 cm ($h = 0.1\text{m}$) and the width is also 8cm ($a = 0.08\text{m}$). For the gravity acceleration, we assume that it equals the standard gravity($g = 9.80665 \text{ m/s}^2$). The speed of sound is also assumed to be 343.2m/s.

Numerical Algorithm

Solutions for Navier-Stokes equations are abundant nowadays. From the previous literature of comparing different methods, we choose the Finite Element Method(FEM) for the numerical approach. FEM is designed to find approximate solutions to partial differential equations and their systems.

In applying FEM, we mainly utilize the toolbox IFISS (Incompressible Flow Iterative Solution Software), designed by D.J. Silvester, H.C. Elman, A. Ramage[2011]. IFISS is a collection of over 290 matlab functions and includes algorithms for discretization by FEM. Within the toolbox, four PDEs are treated: the *Poisson equation*, and steady-state versions of *convection-diffusion equation*, *Stokes equations*, and *Navier-Stokes equations*. Obviously we will use the fourth system. The package allows us to study the effect of different parameters on the contour of the fluid and the estimated error of the analysis.

For the analytic approach, we will plot the graphs of equations (1.11) and (1.14) and observe the change of liquid surface. To implement random vibrations, we randomly add accelerations to our robots to simulate the effect.

Results

First we present the solution of the Navier-Stokes equations using MATLAB toolbox IFISS.

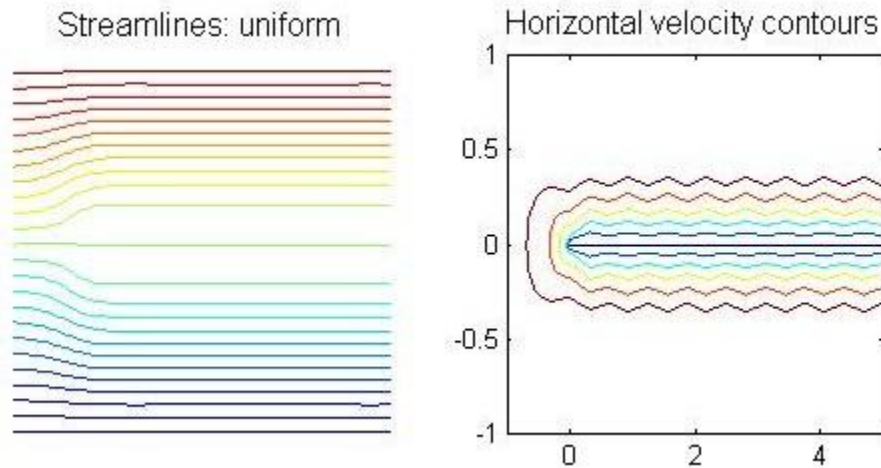


Figure 3 Solution of the Navier-Stokes equations.

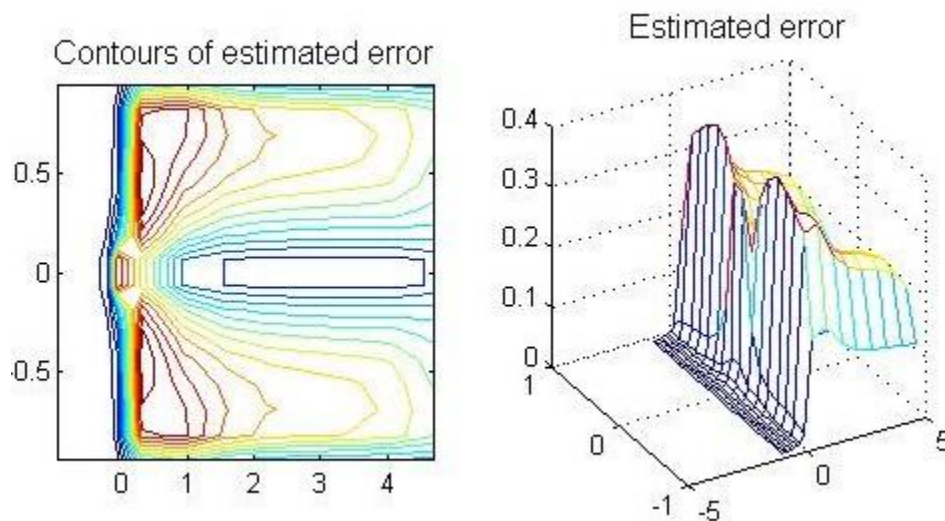


Figure 4 Estimated error of the solution

The liquid horizontal velocity is bouncing around a specific value and estimated error is more concentrated around zero. Therefore, we claim that the solution is valid and acceptable. The back and forth property of liquid motion meets our goal of conducting the simulation.

The core of the model is to plot the graph of surface elevation with respect to time and determine how could control the parameters not to make the liquid spill out of the container. And solving model (1.14) and (1.15) gives us more insight of that. Following is the solution

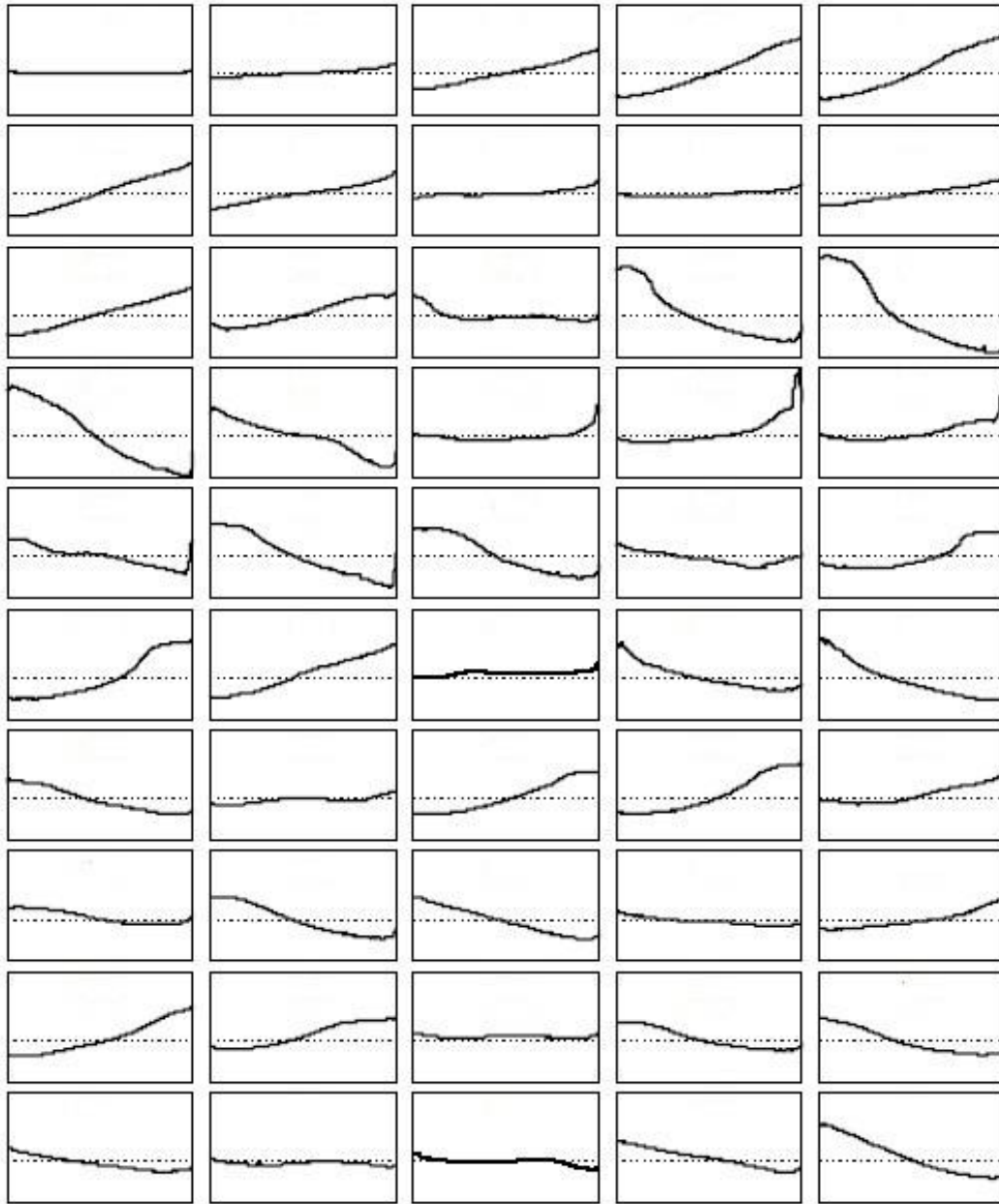


Figure 5 Motion of surface elevation

Factor Analysis

Model (1.11) specifies that parameters $a, \rho, h, \omega, u(t)$ are relative to the surface elevation. Here are our results by using MATLAB.

First we examine the influence of the change in ω . By increasing the angular velocity, surface elevation remains the same as initial condition. Since the acceleration does not change, Nevertheless, its effect on the liquid surface lasts

much longer than the initial condition. This is partly due to larger vorticity, longer the period.

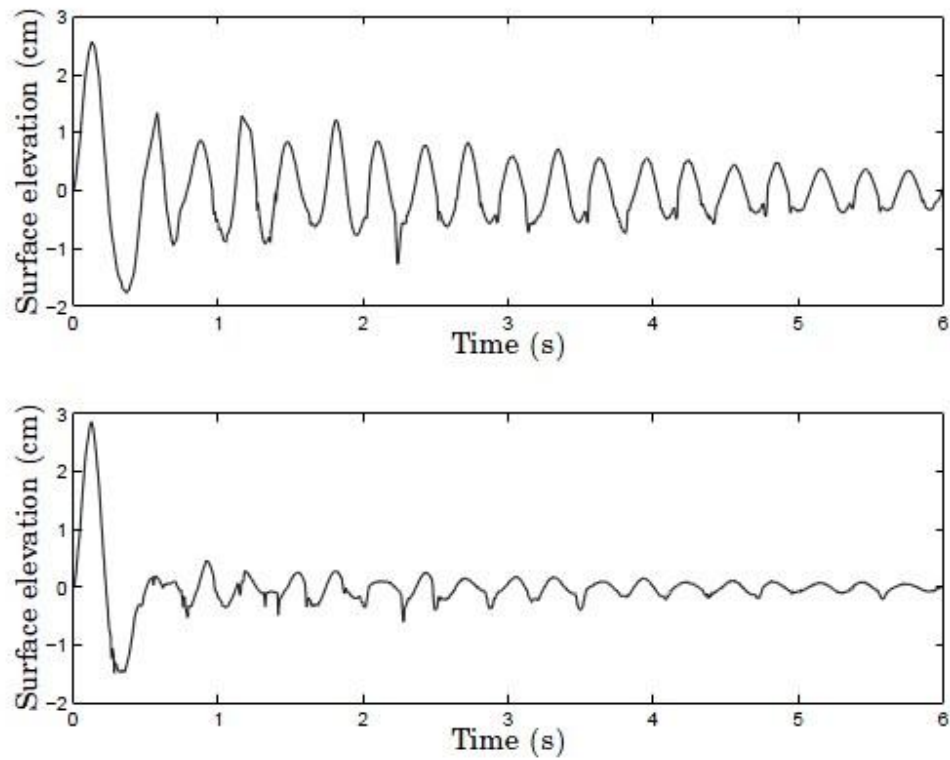


Figure 6 Effect of the change in ω on surface elevation

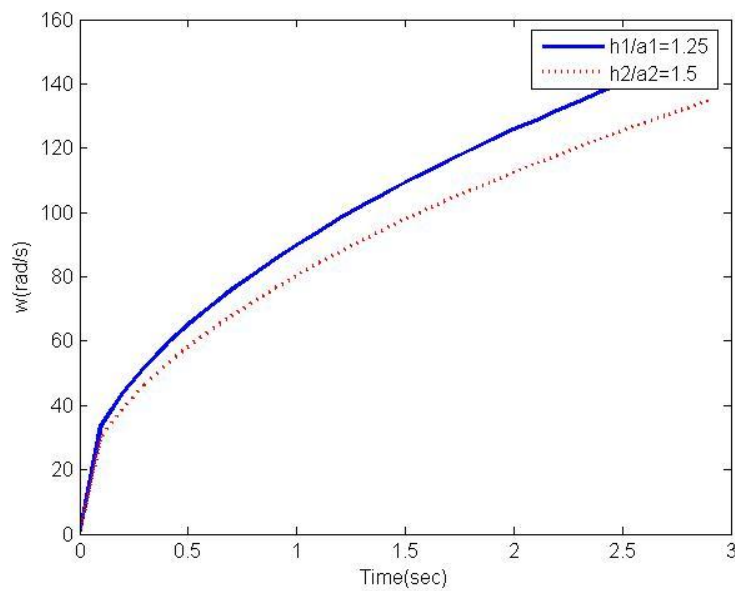


Figure 7 Effect of the change in ratio h/a on ω

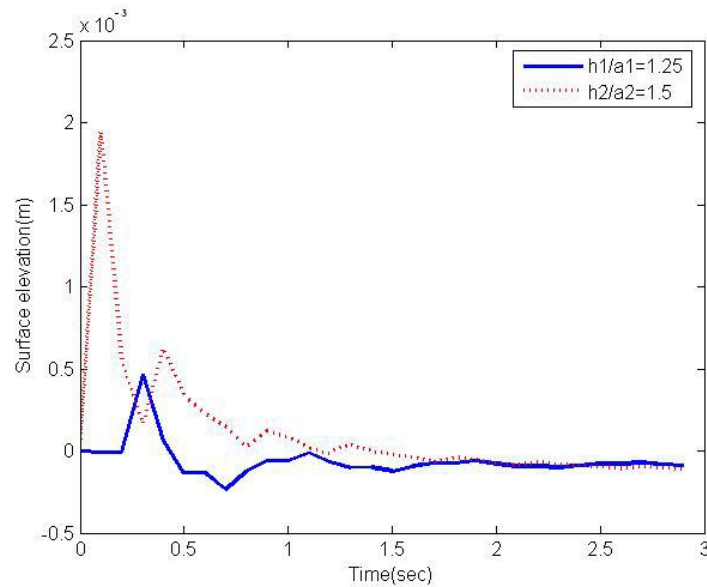


Figure 8 Effect of the change in ratio h/a on the surface elevation

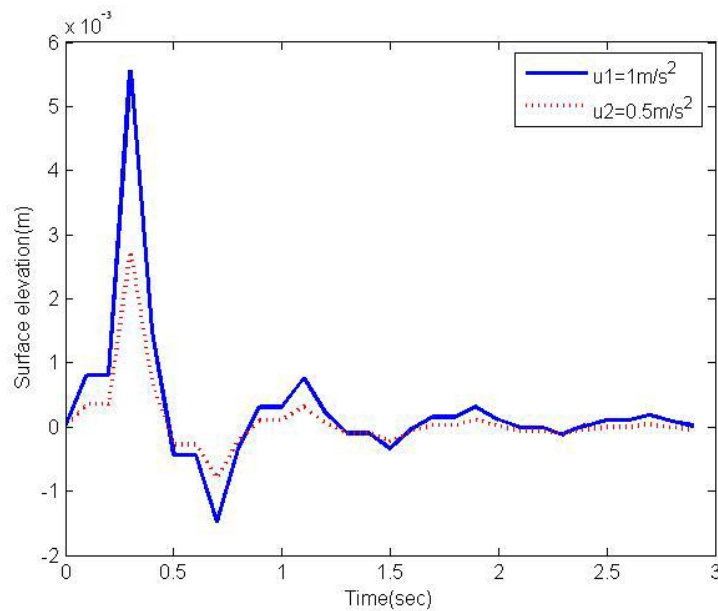


Figure 9 Effect of the change in horizontal acceleration on surface elevation

Since ω is the function of the ratio $\frac{h}{a}$, we would like to see how the change in this ratio influences ω . Figure 7 explains that the change in the ratio does not have a significant effect at the beginning. But later the disparity becomes greater and greater. When the diameters remain the same and depth of liquid larger, ω

seems to be smaller. In other way, when depths of liquid are the same and diameters of container smaller, this would also have the same effect.

For the surface elevation, the change of ratio also appears to be significant according to Figure 8. With higher ratio, surface elevation experiences a surge at the beginning of movement. The patterns of different states are almost the same except the amplitude.

Eventually, we examine how the change in horizontal acceleration affects the surface elevation. The patterns of different states also resemble each other. And with greater acceleration, the liquid surface rises much higher. However, after the first spike, two lines tend to have the same amplitude and pattern.

Task2 Robot Simulation

To simulate the process of walking with a cup of coffee, we theoretically use robot. For the purpose of concentrating on how to avoid coffee spilling, we selectively leave alone the details of the configuration of robot and assume it is in the theoretical perfect condition.

According to daily experience, in most cases, the coffee spills when someone run into you and you have to stop. Therefore, we simulate this effect by commanding robot to stop at time t_* . Here we present the numerical simulation results of this effect.

From Figure 10, we could see that after entrapment, water level rises significantly. Our goal is to control the water not to over the brink of cup. So clearly for this goal, we need to fill in less coffee. But how much is enough? Here we present the numerical result from our model in Figure 11.

To simulate the entrapment, we add an acceleration in an opposite direction. We know from basic physics, object cannot stop immediately if the acceleration becomes zero, only when the velocity is zero. Thus, we implement a relatively larger inverse acceleration to make the robot stop at once, which is also the same process as people do when confronting the situation.

In our model, we only calculate the water level at one side of the cup. From Figure 11, it could be seen that at $t=1.5s$, the surface elevation dives immediately after the impact. It is obvious that if one side is deepen down, the other side of the liquid would enjoy a surge.

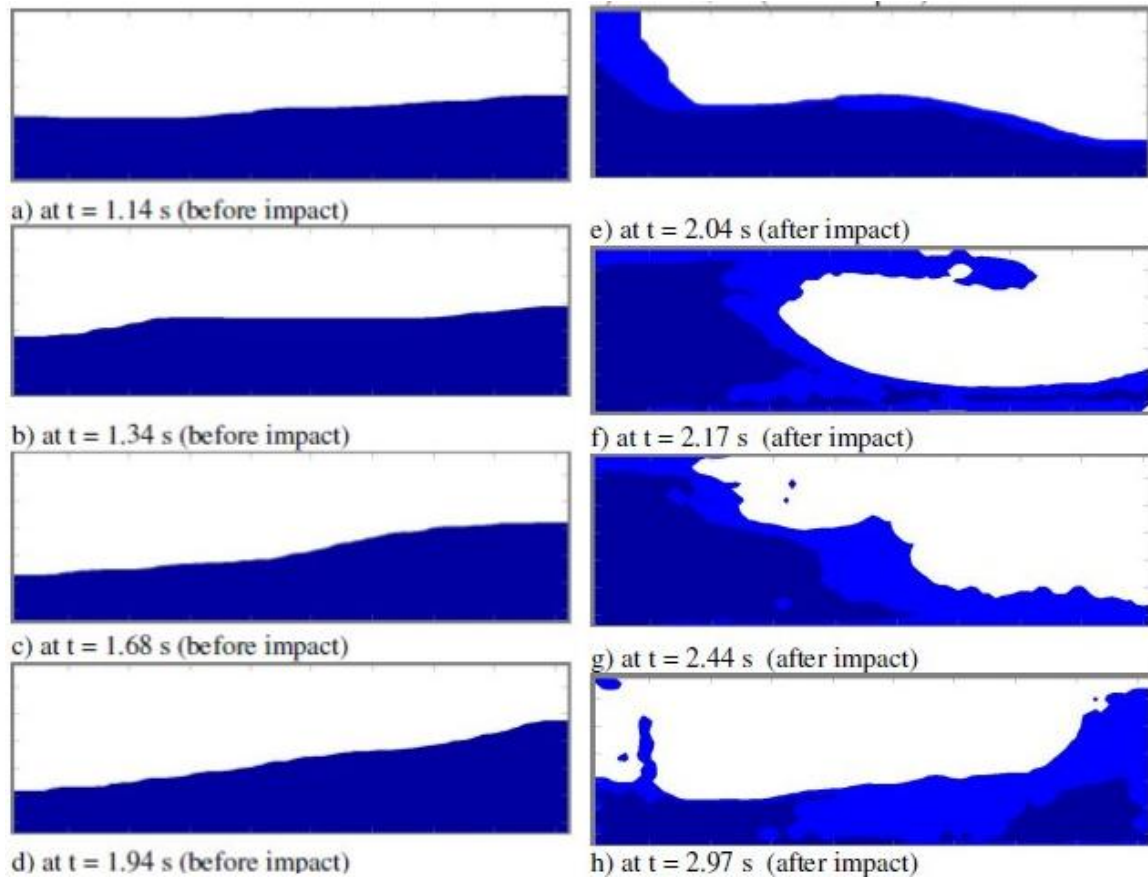


Figure 10 Numerical simulation of entrainment

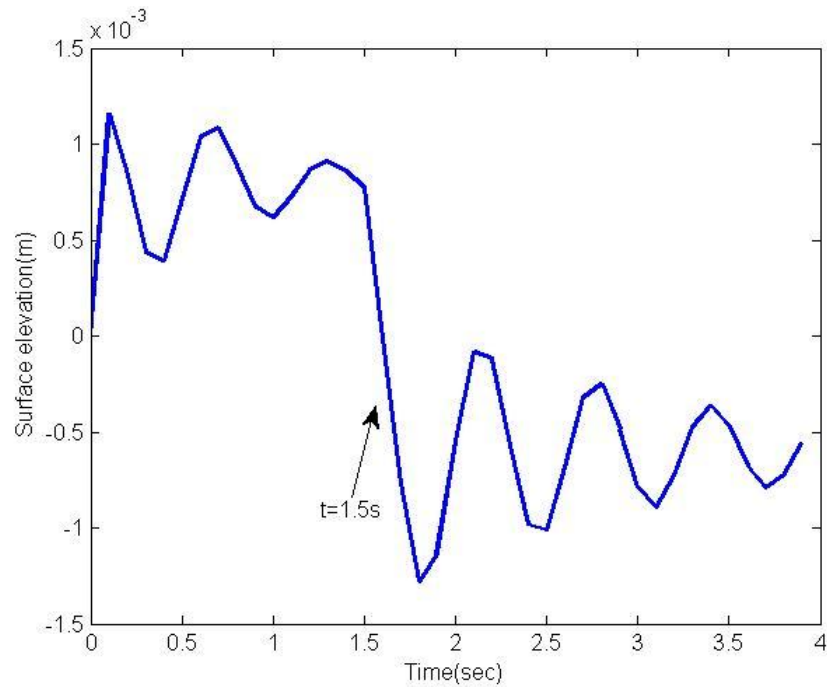


Figure 11 Surface Elevation after the entrainment

Since the surface elevation is only related to the width, depth, vorticity and acceleration and we cannot change the width and depth of the cup at the given situation, the only way to control the coffee not to spill out is to control the acceleration. Nevertheless, due to our rough work in this task, we did not get the exact numerical solution to this problem. But we do know that the strategy is to control the vorticity when robot turns to another direction and the rotational acceleration to avoid the collision.

Conclusions

Whether coffee will spill out of a cup is concerned with the diameters of the cup, depth of coffee in the cup, angular velocity and horizontal acceleration. Greater angular velocity, ratio h/a and horizontal acceleration both lead to higher chance of spilling out of the cup.

For the robot simulation part, after entrapment, the surge of coffee is significant and the strategy for controlling the surface elevation is to control the vorticity and rotational acceleration.

Weakness and Limitations

- Our model is highly theoretical and we believe in empirical experiment, the situation might not match with each other.
- We assume that the viscosity to be zero in model (1.11) to make the calculation easier, which is not prevalent in real situation. Liquid viscosity is usually small but definitely bigger than zero.
- We only consider the horizontal acceleration and eliminate the rotational movements.
- In solving the model, we use the MATLAB toolbox IFISS developed by H.C.ELMAN. IFISS solves Navier Stokes equation in five specific examples, in our case, we choose the option “flow over the plate”, which is the most relative to our model. The example within the algorithm is more specific rather than general. Thus, the solution of our model might not be very precise.
- The values for gravity acceleration and speed of sound are both valid in the standard condition. As basic physics tells, these parameters can vary in different situations.

- As a matter of fact, we didn't get the solution for the random vibration of movement, but only with a more descriptive instead of numerical results. Having descriptive results is less accurate and cogent in simulating the given problem.

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